

OFFICIAL PLATFORM BLUEPRINT & ARCHITECTURE

Institute-wise Research & Consultancy Revenue Monitoring, Awareness & Ranking Platform

A centralized, real-time institutional platform designed to monitor sponsored research, eliminate data silos across all constituent institutes, automate compliance filings for NIRF, NAAC, and QS, and showcase university-wide innovations for public and industrial awareness.

₹4.25 Cr

Initial Annual Target

> ₹5.00 Cr

2030 Revenue Vision

12+

Institutes

Unified Platform

150+

Projects

Industry Target

BASED ON STRATEGIC PROPOSAL BY

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PREPARED BY

Student Engineering & Development Team

Ganpat University • Faculty of Engineering & Technology

1. Executive Summary & Strategic Context

Addressing institutional data fragmentation, awareness gaps, and ranking documentation at Ganpat University

The Current Problem

- **Decentralized Data Silos:** Research grants, consultancy contracts, and student innovations remain isolated across individual laptops, lab notebooks, or departmental files across 12+ institutes.
- **Severe Ranking Audit Penalty:** Every year, university coordinators manually chase faculty for sanction letters, utilization certificates (UC), and audited accounts. Missing documentation leads to underreported research metrics in NIRF and NAAC filings.
- **Zero Campus-Wide Awareness:** Students, research scholars, and junior faculty are unaware of ongoing funded projects, open research openings, and high-end CoE facilities on campus.
- **No Real-time Revenue Tracking:** Constituent institutes lack a live tracking panel to benchmark performance against the university's ₹4.25 Cr annual target.

The Platform Solution

- **Unified University Repository:** A single authoritative database recording all sponsored research, contract R&D, student projects, and IP commercialization.
- **1-Click Ranking Compliance Engine:** Automated export of pre-formatted NIRF Data Capture System (DCS) and NAAC Criterion 3 data sheets with verified document links.
- **Public Showcase & Discovery Portal:** Public-facing searchable catalog of university research projects, patents, and CoE facilities (3D printers, 5G lab) to attract corporate tie-ups.
- **Live Executive CARS Dashboard:** Real-time progress bars tracking institute achievement against designated revenue quotas.

2. National Policy Alignment & 2030 Vision

Fulfilling national research mandates and the CARS 2026–2030 Action Plan

NEP 2020 & ANRF

Aligns with the National Education Policy and Anusandhan National Research Foundation to foster translational, multidisciplinary R&D and extramural funding.

Viksit Bharat 2047

Accelerates domestic technology commercialization, patent generation, and prototype readiness (TRL 1–9) supporting Atmanirbhar Bharat.

Industry-Academia MoUs

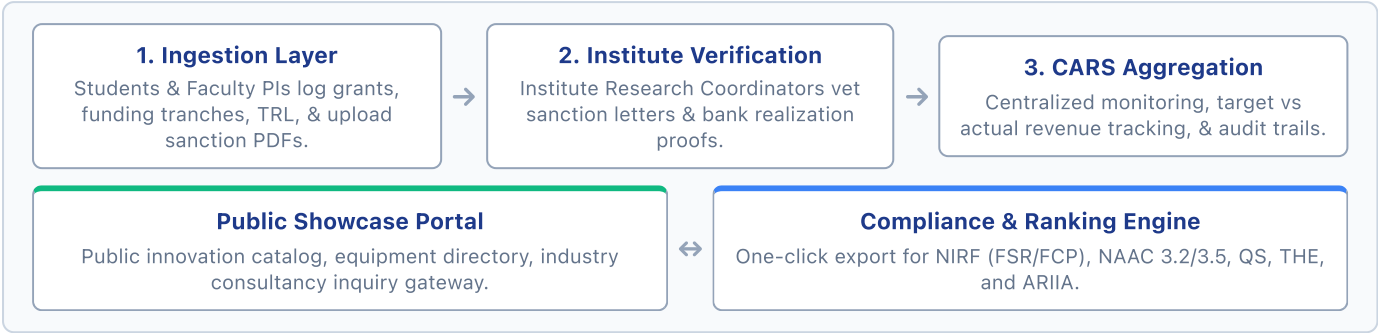
Provides an industry portal for corporate R&D, MSME consulting, and contract manufacturing via GUNI Centres of Excellence.

Expected Outcomes by 2030 (CARS Proposal Target)

- Annual research consultancy revenue exceeding **₹5.00 Crore** by 2030.
- Execution of **150+ industry-sponsored** research projects and **100+ filed/commercialized patents**.
- Active Research Consultancy & Innovation Cells established across **every constituent institute**.
- Position Ganpat University as a preferred research and technical consultancy partner nationally.

3. High-Level Architecture & Data Flow

Multi-tier workflow from researcher entry to executive compliance export



4. Functional Portals & Role-Based Access

Tailored interfaces designed for students, faculty, coordinators, and executive leadership

Public Awareness Gateway (No Login Required)

Audience: Students, Faculty, External Industry & General Public

- Searchable catalog of all active & completed research projects.
- Directory of specialized CoE equipment (e.g. metal 3D printers, 5G IIoT lab, Bosch Rexroth) to invite industry consultancy.
- "Partner with GUNI" inquiry portal for corporate sponsors.

Student & Research Scholar Innovation Portal

Audience: UG/PG Students & Ph.D. Scholars

- Log student innovation grants (SSIP, Hackathons, Incubation funding).
- Discover open research positions posted by faculty PIs.
- Generate verifiable digital portfolio of prototypes, patents, and papers.

Faculty & Principal Investigator (PI) Workspace

Audience: Faculty Members & Principal Investigators

- Guided project intake wizard with auto-suggested funding bodies.
- Installment tracker with Utilization Certificate (UC) upload.
- Faculty incentive and consultancy revenue share calculator.

CARS Executive Dashboard (Dr. Patel & Dr. Gupta)

Audience: CARS Leadership & University Officers

- Real-time revenue monitoring against the ₹4.25 Cr quota.
- 1-Click Ranking Export Engine (NIRF, NAAC, ARIIA).
- Institute-wise performance leaderboard and gap analysis.

5. Comprehensive Data Collection Parameters

Mandatory data attributes specified in Section 5 of the CARS proposal document

A. Project & Personnel Parameters

FIELD NAME	DATA TYPE	DESCRIPTION & ACCEPTED VALUES
Project Title	Text (String)	Full descriptive title of the research consultancy or innovation project
Category / Scope	Dropdown	Sponsored Research, Contract R&D, Industry-funded R&D, Prototype Development, Product Validation, Technology Transfer, Patent Licensing, Student Innovation/SSIP
Funding Agency / Partner	Text + Category	Govt ANRF, DST, DBT, ICAR, ICMR, BIRAC, DRDO, ISRO, MSME, AICTE, CSIR, SERB, GSBTM Industry Corporate R&D, MSME, International, or Internal Seed Grant
Investigators & Team	Relational References	Principal Investigator (PI), Co-PIs (Faculty across GUNI), and Enrolled Students / Ph.D. Scholars (ID, Department, Role)
Institute / Department	Dropdown	Constituent Institute (FOET, FODE, SKPCER, MUIS, DCS/AMPICS, Architecture, CoEs, etc.)
Project Duration & TRL	Date Range + Level	Start Date, Completion Date, and Technology Readiness Level (TRL 1 through TRL 9)

B. Financial & Revenue Tracking Parameters

FINANCIAL METRIC	FORMAT	AUDIT PURPOSE & RANKING LINKAGE
Total Sanctioned Cost	Currency (₹)	Total approved budget stated in the official sanction letter
Sanction Letter Document	File (PDF)	Mandatory Audit Proof required for NIRF/NAAC data verification
Received Revenue (Realized)	Currency (₹)	Actual cash realized in GUNI bank accounts (broken down by installment/tranche)
Utilization Certificate (UC)	File (PDF)	Signed UC uploaded per tranche for financial clearance
Industry Contribution	Currency (₹)	Split into Cash component vs In-Kind value (hardware, tools, software licenses)
Commercial IP Royalties	Currency (₹)	Revenue from patent licensing, technology transfer agreements, or product sales

C. Research Outputs & Intellectual Property (IP)

OUTPUT DIMENSION	KEY ATTRIBUTES	ACCREDITATION & PUBLIC SHOWCASE VALUE
Patents	App No, Filing/Grant Date	Published and granted patents feed directly into NAAC 3.4.3 & NIRF RPC
Prototypes & Technologies	Name, Photos, TRL	Showcased on the public portal to demonstrate industrial readiness
Startups Incubated	Startup Name, CIN, Team	Links student & faculty ventures to GUNI Incubation / ARIIA reporting
Publications & MoUs	DOI, WoS/Scopus ID	Indexed papers with impact factors; active corporate MoUs

6. Institute-wise Annual Consultancy Revenue Targets

Official revenue quotas established by CARS to reach ₹4.25 Cr annually (Scaling to > ₹5.00 Cr by 2030)

CONSTITUENT INSTITUTE / SCHOOL	TARGET (₹ CR)	PRIMARY CONSULTANCY & RESEARCH FOCUS AREAS
Faculty of Engineering & Technology (FOET / UVPCE / ICT)	₹ 1.30 Cr	AI, Robotics, Industry 4.0, Renewable Energy, Automobile, Semiconductors, Testing
Centres of Excellence (Additive Mfg, 5G, Bosch Rexroth)	₹ 1.00 Cr	Metal/Polymer 3D Printing, Cobots, 5G Industrial IoT, Digital Twin, Defence/Aerospace
Faculty of Science (MUIS)	₹ 0.30 Cr	Biotech, Material Characterization, Chemical Analysis, Environmental Studies
Faculty of Diploma Engineering (FODE)	₹ 0.30 Cr	Industrial Testing, SME Prototyping, CAD/CAM/CAE Simulation Services
Faculty of Agriculture Allied Sciences & Technology	₹ 0.25 Cr	Agri-Tech, Precision Farming, Food Processing/Packaging, Seed Biotech
Faculty of Computer Applications (DCS / AMPICS)	₹ 0.20 Cr	FinTech, AI Solutions, Cybersecurity Audits for SMEs, Digital Transformation
Faculty of Pharmacy (IOPH / SKPCER)	₹ 0.20 Cr	Drug Formulation, Pharmacovigilance, GMP Compliance Consultancy, Testing
Faculty of Architecture, Design & Planning	₹ 0.20 Cr	Smart City Advisory, Green Building, GIS & Spatial Analytics, BIM, Heritage
Faculty of Health & Allied Sciences (Physiotherapy)	₹ 0.15 Cr	Ergonomic Assessments, Rehabilitation Planning, Sports Performance Analytics
Nursing	₹ 0.15 Cr	Community Health Consultancy, Occupational Industrial Health Programs
Faculty of Management Studies (VMPIM / CMSR / VMPCIMS)	₹ 0.10 Cr	MSME Advisory, Startup Incubation Mentorship, Executive Development
Social Science & Humanities	₹ 0.10 Cr	Policy Development, Social Impact Assessments, Urban Socio-Economic Studies
TOTAL ANNUAL REVENUE TARGET	₹ 4.25 Cr	Official Target Established by CARS Leadership

7. Core Centre of Excellence (CoE) Assets for Commercialization

High-end facilities to be cataloged on the public portal to drive corporate consultancy contracts

Additive Manufacturing & Smart Systems

- Advanced 3D printers (metal and polymer sintering).
- 5-axis CNC machines and smart manufacturing lines.
- Reverse engineering and rapid prototyping scanners.

5G, Automation & Industry 4.0

- 5G industrial communications and Industrial IoT (IIoT) testbeds.
- Bosch Rexroth hydraulic, pneumatic, and PLC/SCADA benches.
- Collaborative robots (Cobots) & Digital Twin / AR-VR facilities.

8. Direct University Ranking & Filing Mechanics

Automating annual audit submissions from continuous real-time data collection

NIRF (Research & Professional Practice - RPC)

- FSR (Funded Sponsored Research):** Generates verified tables of government and industry projects with audited sanctioned amounts.
- FCP (Funded Consultancy Projects):** Auto-calculates exact client consultancy cash receipts per financial year with downloadable proof attachments.

NAAC (Criterion 3: Research & Innovations)

- Metric 3.2.1 / 3.2.2:** Extramural research funding data pre-formatted into official NAAC Excel templates.
- Metric 3.4.3 / 3.4.4:** Complete patent registry (published and granted).
- Metric 3.5.1 / 3.5.2:** Audited consultancy revenue and corporate training fees.

QS & Times Higher Education (THE)

- Industry Income / Knowledge Transfer:** Instant calculation of research revenue generated from industry per academic staff.

ARIIA (Innovation Ranking)

- Tracks progression of Technology Readiness Levels (TRLs), working prototypes, student startups, and IP commercialization royalties.

9. Proposed 4-Sprint Implementation Roadmap

Structured development timeline for the student development team

SPRINT	FOCUS AREA	DELIVERABLES & MILESTONES
Sprint 1 (W1-2)	Schema & Intake Forms	Relational database schema for projects, tranches, and IP; multi-role authentication; project intake forms with PDF upload.
Sprint 2 (W3-4)	Dashboards & Quotas	Institute target progress bars (₹4.25 Cr matrix); institute coordinator review and approval workflow.
Sprint 3 (W5-6)	Ranking Export Engine	Automated Excel export generator formatted specifically for NIRF (RPC) and NAAC (Criterion 3.2 & 3.5).
Sprint 4 (W7-8)	Public Showcase & Pilot	Public searchable innovation and CoE equipment directory; pilot testing with FOET and Centres of Excellence.

Next Action for Student Team

Present this blueprint directly to **Dr. Heena Patel** and **Dr. Ajay Kumar Gupta** to validate the data schema, confirm the approval workflows, and obtain sign-off to begin active development.